I worked on a real business problem in demand forecasting. Traditional process is highly manual, time-consuming, and heavily dependent on planners, which made it difficult to scale and maintain consistency.

As part of my research, I conducted both quantitative and qualitative analysis comparing traditional forecasting methods with machine learning approaches. The findings showed that neither approach alone was sufficient — traditional methods lacked scalability, while ML models struggled with high variability and business context. Based on this, I proposed a hybrid approach that combines the strengths of both techniques in a structured way.

Low-impact or low-volume items can be effectively handled using ML models like XGBoost.

High-impact or high-volume items still require planner expertise to incorporate business insights, promotions, and market knowledge.

I developed a 4-step demand forecasting process:

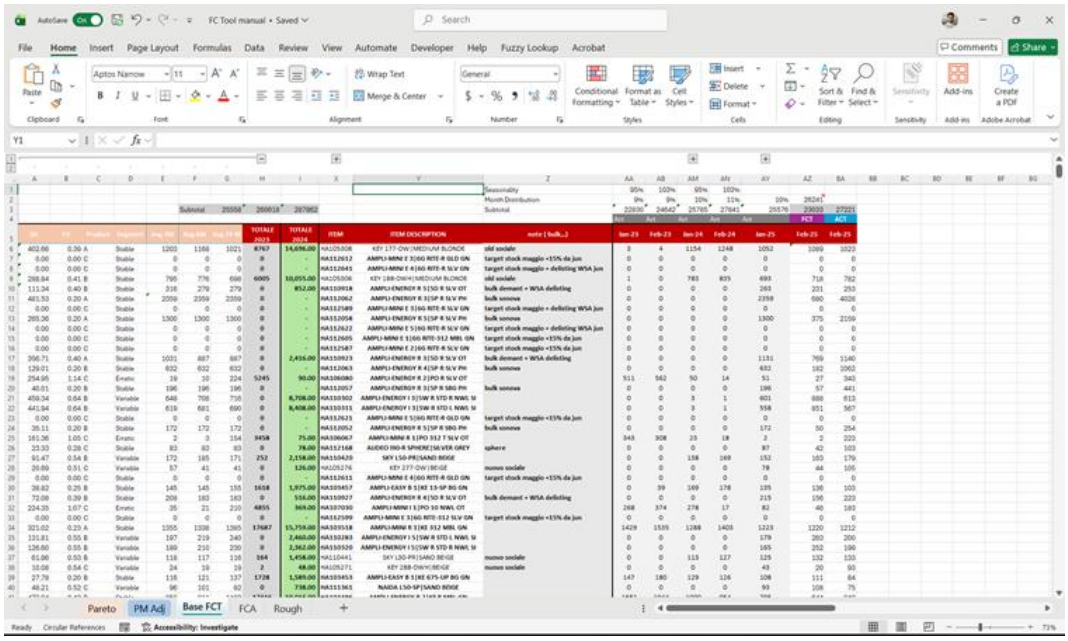
- **ML Forecasting** – Automated forecasts generated for all SKUs
- **Excel-Based Forecasting** – Traditional statistical forecasts for baseline comparison
- **Forecast Combination** – Averaging ML and traditional forecasts to improve stability
- **Planner Adjustment (Pareto-Based)** – Planners focus only on high-impact items using Pareto analysis and S&OP inputs.

Impact:

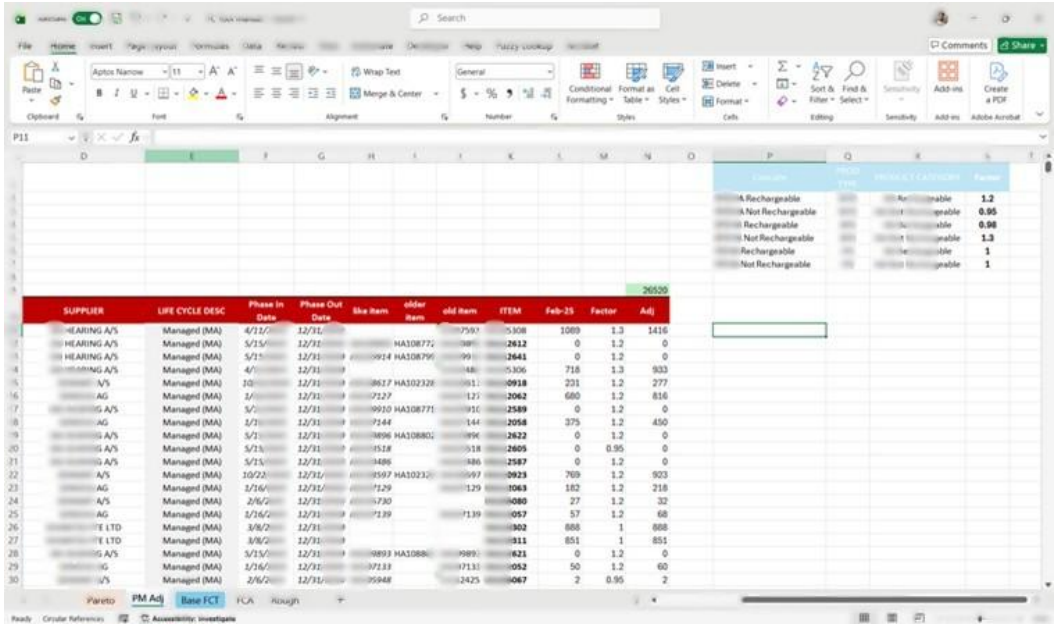
- Reduced planner workload from 1000+ SKUs to around 100 key items
- Improved forecasting accuracy by approximately 4%
- Enabled planners to shift from manual work to strategic decision-making and exception management

Due to data privacy and confidentiality constraints, sensitive information has been blurred.

Excel-Based Forecasting :



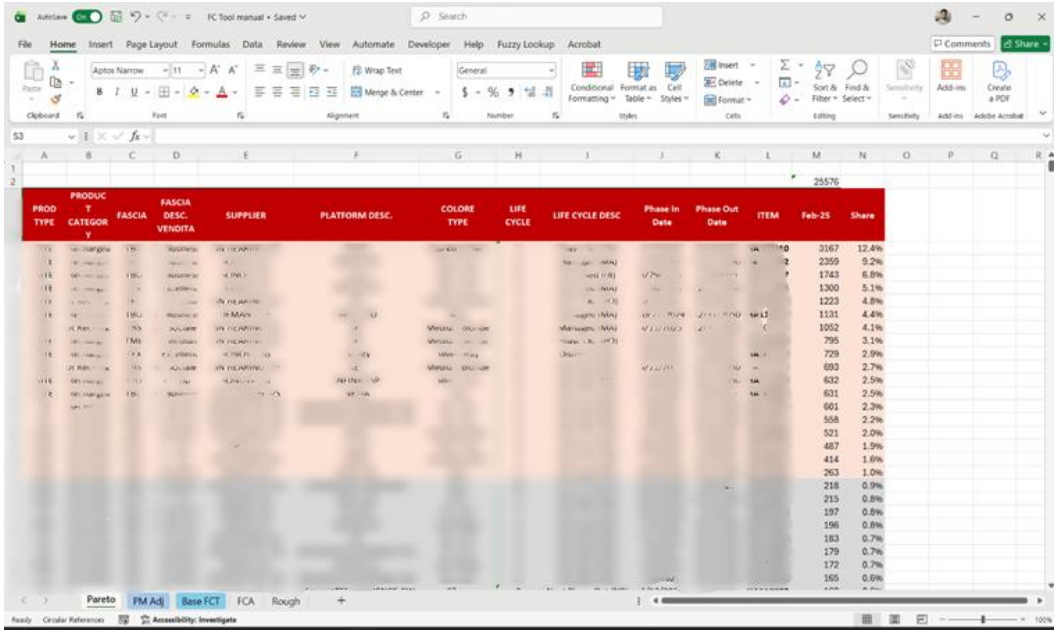
Planner Adjustment :



SUPPLIER	LIFE CYCLE DESC	Phase In Date	Phase Out Date	like item	older item	old item	ITEM	Feb-25	Factor	Adj
HEARING A/S	Managed (MA)	4/11/	12/31/			7593	5308	1089	1.3	1416
HEARING A/S	Managed (MA)	5/15/	12/31/		HA108772		2612	0	1.2	0
HEARING A/S	Managed (MA)	1/15/	12/31/		9914 HA108799		2641	0	1.2	0
HEARING A/S	Managed (MA)	4/	12/31/				5306	718	1.3	933
A/S	Managed (MA)	10/	12/31/		8617 HA102328		0918	231	1.2	277
AG	Managed (MA)	1/	12/31/		7127		2062	680	1.2	816
G A/S	Managed (MA)	5/	12/31/		9910 HA108771		2589	0	1.2	0
AG	Managed (MA)	1/1	12/31/		7144		2058	375	1.2	450
G A/S	Managed (MA)	5/1	12/31/		3496 HA108802		2622	0	1.2	0
G A/S	Managed (MA)	5/15	12/31/		3518		2605	0	0.95	0
G A/S	Managed (MA)	5/15	12/31/		3486		2587	0	1.2	0
A/S	Managed (MA)	10/22	12/31/		3597 HA102332		0923	769	1.2	923
AG	Managed (MA)	1/16/	12/31/		7129		1063	182	1.2	218
A/S	Managed (MA)	2/6/2	12/31/		5730		0080	27	1.2	32
AG	Managed (MA)	1/16/2	12/31/		7139	7139	057	57	1.2	68
TE LTD	Managed (MA)	3/8/2	12/31/				302	888	1	888
TE LTD	Managed (MA)	3/8/2	12/31/				311	851	1	851
G A/S	Managed (MA)	5/15/	12/31/		9893 HA10886	9893	621	0	1.2	0
G	Managed (MA)	1/16/	12/31/		7133	7133	052	50	1.2	60
A/S	Managed (MA)	2/6/2	12/31/		5948	2425	067	2	0.95	2

A Rechargeable	1.2
A Not Rechargeable	0.95
Rechargeable	0.98
Not Rechargeable	1.3
Rechargeable	1
Not Rechargeable	1

Pareto Analysis:



PROD TYPE	PRODUCT CATEGORY	FASCIA	FASCIA DESC. VENDITA	SUPPLIER	PLATFORM DESC.	COLORE TYPE	LIFE CYCLE	LIFE CYCLE DESC	Phase In Date	Phase Out Date	ITEM	Feb-25	Share
												3167	12.4%
												2359	9.2%
												1743	6.8%
												1300	5.1%
												1223	4.8%
												1131	4.4%
												1052	4.1%
												795	3.1%
												729	2.9%
												693	2.7%
												632	2.5%
												631	2.5%
												601	2.3%
												558	2.2%
												521	2.0%
												487	1.9%
												414	1.6%
												263	1.0%
												218	0.9%
												215	0.8%
												197	0.8%
												196	0.8%
												183	0.7%
												179	0.7%
												172	0.7%
												165	0.6%